Fall 2026 ECE 255: Introduction to Computer Architecture (Units: 1.5)

Territory Acknowledgement

We acknowledge and respect the Lək̓ʷəŋən (Songhees and Xʷsepsəm/Esquimalt) Peoples on whose territory the university stands, and the Lək̓ʷəŋən and W̱SÁNEĆ Peoples whose historical relationships with the land continue to this day.

Key Dates and Schedules

Section	CRN	Schedule	Location	Class Start	Class End	Days of Weeks*	Hours of Day	Instructor
A01	10988	Lecture	BWC A104	2026-09-09	2026-12-07	TWF	09:30-10:20	Kin Fun Li
B01	10989	Lab	ELW B328	2026-09-28	2026-12-04	M	13:30-16:20	Kin Fun Li
B02	10990	Lab	ELW B328	2026-09-28	2026-12-04	T	13:30-16:20	Kin Fun Li
B03	10991	Lab	ELW B328	2026-09-28	2026-12-04	W	16:30-19:20	Kin Fun Li
B05	10992	Lab	ELW B328	2026-09-28	2026-12-04	T	16:30-19:20	Kin Fun Li

*M=Monday, T=Tuesday, W=Wednesday, R=Thursday, F=Friday.

Withdrawal without reduction of tuition fees: 2026-09-22
Withdrawal with 50% reduction of tuition fees: 2026-10-13
Last day for withdrawal (no fees returned): 2026-10-31

Instructor(s), Contact Information, and Office Hours

Name: **Kin Fun Li**
Office: EOW 409
Phone: 250 721 8683
Email: kinli@uvic.ca

Office Hours: Tuesdays and Wednesdays 10:30 - 11:30 a.m.

TA Information

TBA

Pre- & Co-Requisites

Prerequisites

- Complete 1 of:
  - CSC110 - Fundamentals of Programming I (1.5)
  - CSC111 - Fundamentals of Programming with Engineering Applications (1.5)

Pre- or corequisites

- Completed or concurrently enrolled in 1 of:
  - CSC115 - Fundamentals of Programming II (1.5)
  - CSC116 - Fundamentals of Programming with Engineering Applications II (1.5)

Course Objectives

To gain an understanding of how a computer system works and its subsystems interact.

To familiarize the control of low-level computer operations using assembly language programming.

Learning Outcomes

- Able to identify suitable computer hardware and system software for specific engineering applications.
- Able to appreciate the synergy of computer system hardware and software.
- Able to relate high-level algorithmic concepts and programming languages to machine-level system hardware and software.

Syllabus

- The architecture of computer systems including concepts such as processor, memory, buses, input/output, instruction sets, interrupt processing, pipelining, performance
- Families of processors, CISC, RISC
- Memory organization and management including cache and virtual memory
- Computer arithmetic
- Assembly language programming, assemblers, linkers and loaders
- Hardware/Software interaction

Textbook & Course Materials

Lecture Notes
Available on Course Brightspace

OPTIONAL TEXT	
Title:	Computer Organization & Embedded Systems (6)
Author:	Hamacher, Vranesic, Zaky, & Manjikian
Publisher/Year:	McGraw Hill, 2012

Laboratory Manual	
Title:	ECE 255 Laboratory Manual
Author:	KFL et al.
Publisher / Year:	Available on Course Brightspace

Course Delivery

Lectures are conducted live.

Laboratories are conducted live.

Quizzes are conducted live, in class via Brightspace.

Office hours are conducted in Zoom; in-person appointment can be arranged via email (kinli@uvic.ca)

Learning & Teaching Technologies

- Desktop or Laptop Computer with webcam/microphone/speakers/headsets for office hours & tutorials.
- Web browser to be able to access/run Zoom, Brightspace, etc.
- PDF viewer.
- Excel, Word, etc. might be useful as well.
- Please make yourself familiar with Zoom, which we will use for live interaction in tutorial, office hours, etc.

Assessment

Assessment	Weight	Date
Labs	25%	According to lab section schedule
Test #1	15%	September 29, 2026
Test #2	20%	October 21, 2026
Test #3	20%	November 18, 2026
Test #4	15%	December 4, 2026
Quizzes (5)	5%	In class, random dates

Notes

Failure to complete all laboratory requirements, including in-lab work and lab reports, will result in a grade of N being awarded for the course.

The 4 tests throughout the term count for 70% of the course grade. To account for an academic concession request for a missed test, the grade percentage of the other 3 tests in total will count as the missed test grade percentage. No other concession will be granted for missed tests.

A missed in-class quiz will be awarded a 0.

A supplemental exam will not be offered in this course.

The final grade obtained from the above marking scheme for GPA calculation will be based on the percentage-to-grade point conversion table listed in the current Undergraduate Calendar.

Coursework Mark Appeals: All marks must be appealed within 7 days of the mark being posted.

Regarding the use of Generative AI:

Using artificial intelligence tools is strictly prohibited for completing any labs, reports, quizzes, or exams in this course. All submitted work must be your own original creation. Using AI for your work will be treated as an academic integrity violation.

Providing Feedback

Students can provide in-term feedback to the instructor via email (kinli@uvic.ca) or in-person appointment.

General Information

Note to students: Students who have issues with the conduct of the course should discuss them with the instructor first. If these discussions do not resolve the issue, then students should feel free to contact the Chair of the Department by email, or the Chair's Assistant to set up an appointment.

Course Lecture Notes

Unless otherwise noted, all course materials supplied to students in this course have been prepared by the instructor and are intended for use in this course only. These materials are not to be recirculated, whether by emailing, uploading, copying to websites, or to others not enrolled in this course. Violation of this policy may in some cases constitute a breach of academic integrity as defined in the UVic Calendar.

Academic Integrity

Academic Integrity requires commitment to the values of honesty, trust, fairness, respect, and responsibility. It is expected that students, faculty members and staff at the University of Victoria, as members of an intellectual community, will adhere to these ethical values in all activities related to learning, teaching, research and service. Any action that contravenes this standard, including misrepresentation, falsification or deception, undermines the intention and worth of scholarly work and violates the fundamental academic rights of members of our community. This policy is designed to ensure that the university's standards are upheld in a fair and transparent fashion.

Attendance

Students are expected to attend all classes in which they are enrolled. An academic unit may require a student to withdraw from a course if the student is registered in another course that occurs at the same time.

- An instructor may refuse a student admission to a lecture, laboratory, online course discussion or learning activity, tutorial or other learning activity set out in the course outline because of lateness, misconduct, inattention or failure to meet the responsibilities of the course set out in the course outline. Students who neglect their academic work may be assigned a final grade of N or debarred from final examinations.
- Students who do not attend classes must not assume that they have been dropped from the course by an academic unit or an instructor. Courses that are not formally dropped will be given a failing grade; students may be required to withdraw and will be required to pay the tuition fee for the course.

See further information on attendance in the Undergraduate Academic Calendar.

Centre for Accessible Learning (CAL)

The University of Victoria is committed to creating a learning experience that is as accessible as possible. If you are registered with the Centre for Accessible Learning and anticipate or experience any barriers to learning in this course, please feel welcome to discuss your concerns with me. If you are a student with a disability or chronic health condition, you can meet with a CAL advisor to discuss access and accommodations.

Academic Concession

The University recognizes its responsibility to offer academic concessions to students whose ability to complete course requirements is interrupted by unexpected and unavoidable circumstances or conflicting responsibilities. Review the Academic Concession Regulation in the Undergraduate Academic Calendar and website.

Grading System

The University of Victoria follows a percentage grading system in which the instructor will submit grades in percentages. The University will use the following Senate approved standardized grading scale to assign letter grades. Both the percentage mark and the letter grade will be recorded on the academic record and transcripts.

F	D	C	C+	B-	B	B+	A-	A	A+
0-49	50-59	60-64	65-69	70-72	73-76	77-79	80-84	85-89	90-100

Grades	Description
A+, A, A-	**Exceptional**, **outstanding** or **excellent** performance. Normally achieved by a minority of students. These grades indicate a student who is *self-initiating*, *exceeds expectation* and has an *insightful* grasp of the subject matter.
B+, B, B-	**Very good**, **good** or **solid** performance. Normally achieved by the largest number of students. These grades indicate a *good* grasp of the subject matter or *excellent grasp in one area balanced with satisfactory grasp in the other areas*.
C+, C	**Satisfactory**, or **minimally satisfactory**. These grades indicate a *satisfactory performance and knowledge* of the subject matter.
D	**Marginal Performance**. A student receiving this grade demonstrated a *superficial grasp* of the subject matter.
F	**Unsatisfactory performance**. Wrote final examination and completed course requirements; no supplemental.

Student Experience of Learning (SEL)

I value your feedback on this course. Towards the end of term you will have the opportunity to complete a confidential Student Experience of Learning (SEL) survey regarding your learning experience. The survey is vital to providing feedback to me regarding the course and my teaching, as well as to help the department improve the overall program for students in the future. When it is time for you to complete the survey, you will receive an email inviting you to do so. If you do not receive an email invitation, you can go directly to the SEL site

You will need to use your UVic NetLink ID to access the survey, which can be done on your laptop, tablet or mobile device. I will remind you closer to the time, but please be thinking about this important activity, especially the following three questions, during the course.

- What strengths did your instructor demonstrate that helped you learn in this course?
- Please provide specific suggestions as to how the instructor could have helped you learn more effectively.
- Please provide specific suggestions as to how this course could be improved.

Equality

This course aims to provide equal opportunities and access for all students to enjoy the benefits and privileges of the class and its curriculum and to meet the syllabus requirements. Reasonable and appropriate accommodation will be made available to students with documented disabilities (physical, mental, learning) in order to give them the opportunity to successfully meet the essential requirements of the course. The accommodation will not alter academic standards or learning outcomes, although the student may be allowed to demonstrate knowledge and skills in a different way. It is not necessary for you to reveal your disability and/or confidential medical information to the course instructor. If you believe that you may require accommodation, the course instructor can provide you with information about confidential resources on campus that can assist you in arranging for appropriate accommodation. Alternatively, you may want to contact the Centre for Accessible Learning located in the Campus Services Building.

The University of Victoria is committed to promoting, providing, and protecting a positive, and supportive and safe learning and working environment for all its members.

Resources for students

a. UVicLearn is the primary learning resource for students that offers many learning workshops and resources to help students with academics and learning strategies.
b. Library resources
c. Indigenous Student Services (ISS)
d. Academic Skills Centre (ASC)
e. Math & Stats Assistance Centre (MSAC)
f. Learning Assistance Program (LAP)
g. Community-Engaged Learning (CEL)
h. Learning in Motion - Co-op and Career Portal
i. Academic concession
j. Academic accommodation
k. Academic accommodation & access for students with disabilities
l. Student groups and resources
m. Student wellness
n. Ombudsperson

University statements and policies

a. University Calendar - Information for all students
b. Accommodation of Religious Observance
c. Student Conduct
d. Non-academic Student Misconduct
e. Academic Accommodations and Accessibility
f. Diversity / EDI – VPAC commitment
g. Sexualized Violence Prevention and Response
h. Discrimination and Harassment

Copyright Statement

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